Kaiven Zhou

Senior Data Engineer

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SUMMARY

Senior Data Engineer with 9+ years of experience building scalable data platforms and production data pipelines at DoorDash, Facebook, and Microsoft. I specialize in designing reliable ETL and streaming systems using Python, SQL, Spark, Snowflake, and AWS to transform large-scale event data into trusted analytical and machine learning datasets. My work focuses on fraud detection, experimentation infrastructure, and platform observability, where I partner closely with product, data science, and security teams to deliver high-quality data systems that power real-time decisions and measurable business impact.

EXPERIENCE

Senior Data Engineer

DoorDash 08/2021 - Present

- Spearheaded data engineering for project FraudShield, building scalable data pipelines and curated fraud datasets, supporting detection anomaly discovery and financial loss measurement across consumer, merchant, and dasher transactions.
- Built transaction-level ingestion pipeline using Python, AWS S3, and Snowflake to normalize payment authorization events, enabling analysts to trace chargeback patterns and reduce investigation query latency for fraud operations.
- Developed ELT workflows orchestrated with Airflow and dbt, transforming marketplace events into dimensional fraud mart tables in Amazon Redshift, improving SQL-based risk reporting for trust and safety analysts.
- Implemented large-scale feature generation jobs with PySpark on Databricks, writing Delta Lake tables that produced model-ready behavioral aggregates used by fraud scientists to train card abuse detection models.
- Designed streaming ingestion for order and payment events using Kafka and Python services, enabling near real-time fraud signals to reach risk scoring APIs within seconds for transaction blocking.
- Established CI/CD pipelines with GitHub, Docker, and Kubernetes, deploying data pipeline services and dbt models automatically, which improved release reliability and reduced manual schema migration errors across fraud datasets.
- Implemented data quality monitoring using SQL validation checks and warehouse alerts in Snowflake, detecting schema drift and missing partitions early and protecting downstream fraud dashboards used by operations teams.
- Partnered with security and compliance teams to implement role-based access controls and audit logging across AWS data lake resources, ensuring sensitive payment and identity datasets met governance requirements.

SKILLS

Language & Scripting

Python · SQL · Scala · Java · Bash · TypeScript · JavaScript

Data Engineering & Analytics Tools

Apache Spark · PySpark · Databricks · dbt · Apache Airflow · ETL/ELT Pipelines · Data Modeling · Data Warehousing · Snowflake · Amazon Redshift · Delta Lake · Kafka · REST APIs · NoSQL Databases

Cloud & Infrastructure

AWS · Azure · Docker · Kubernetes · Terraform · CI/CD Pipelines · Git

Visualization & Monitoring

Power BI · Tableau · Alerting Systems · Data Quality Monitoring · Pipeline Observability

EXPERIENCE

Data Engineer

Facebook 01/2019 - 07/2020

- Engineered data infrastructure for project ExperimentLens, improving the reliability of experiment exposure and outcome datasets that powered company-wide product A/B testing and decision making.
- Built analytical tables in Hive and optimized Presto SQL queries that computed experiment metrics from exposure and conversion logs, enabling faster experiment readouts for product scientists.
- Developed Python-based data pipelines scheduled with Airflow to transform streaming exposure logs into partitioned warehouse datasets, improving the freshness and consistency of experimentation metrics used across product teams.
- Implemented automated data validation and anomaly alerts using SQL checks and dashboards, allowing engineers to detect metric regressions in experiment pipelines before results were consumed by decision makers.
- Created internal REST APIs serving curated experiment datasets, enabling downstream analytics tools and notebooks to access consistent metrics without repeatedly querying large raw logging tables.

Data Engineer

Microsoft 07/2017 - 12/2018

- Worked on project InfraMetrics Platform, building data collection and aggregation systems that tracked service requests latency and infrastructure health across Yammer production environments globally supporting reliability.
- Implemented Python ETL jobs, extracting operational metrics from Postgres and service logs then loading structured datasets for reliability dashboards used by infrastructure engineers during incident investigations.
- Built distributed processing workflows with Apache Spark on Hadoop clusters to aggregate high-volume request traces, producing daily latency summaries that guided capacity planning and performance tuning.
- Introduced CI/CD automation with Git and Docker for infrastructure data services, improving deployment consistency and enabling faster iteration on telemetry processing components across environments reliably.
- Developed data access services integrating Cassandra operational stores with analytics pipelines, enabling engineers to query historical request behavior and debug production incidents more effectively during outages.

EDUCATION

BS, Operations Research

Columbia University 2011 - 2015

SKILLS

Others

- Streaming Data Processing
- Event-Driven Architectures
- Data Governance
- Security & Compliance
- Feature Engineering for Machine Learning

STRENGTHS

Communication

Able to translate complex data platform architecture and analytical findings into clear explanations for product managers, analysts, and engineers, enabling faster decision making and cross-team alignment.

Problem Solving

Enjoy tackling complex data engineering challenges and designing scalable pipelines and data models that turn messy, high-volume event data into reliable and actionable datasets.

Ownership

Focused on delivering practical solutions that improve data reliability, performance, and usability, while ensuring projects move forward and teams stay aligned on measurable outcomes.